HireLucent

Software Requirement Specification

AI-Powered Recruitment and Interview Intelligence Platform

Bringing clarity and intelligence to candidate evaluation.

Version ID Date: 27/10/2026

Document ID: SWD/HL/SRS/001

Version ID: 1.0

Status: Final, for submission

Prepared by: Shahriar Farhan Karim, AI Trainee Developer

Programme: BJIT Academy, AI Trainee Developer Programme

Project: Final Evaluation Assignment, Phase 2

Revision History

Doc. Ver.	Date	Author	Reviewer	Description of Revision
1.0	27/10/2026	Shahriar Farhan Karim	Jishnu Sardar, Md. Minhaz Ul Karim	Initial release: complete requirements of HireLucent as delivered, covering Core features F1–F6, Stretch features F7–F10 and the Innovation feature F11.

Document Approval

The signatures below confirm that this specification has been reviewed and that it is the agreed baseline for the delivered HireLucent system.

Role	Name	Signature	Date
Prepared by	Shahriar Farhan Karim (AI Trainee Developer)		27/10/2026
Reviewed by			
Approved by			

Contents

1. Introduction....4
1.1 Purpose....4
1.2 Scope....4
1.3 Intended Stakeholder....5
1.4 References....5
1.5 Definitions, Acronyms, and Abbreviations....6
2. Overall Description....7
2.1 Overview....7
2.1.1 Product Perspective....7
2.1.2 Product Functions....7
2.1.3 User Characteristics....8
2.1.4 Constraints....8
2.1.5 Assumptions and Dependencies....8
2.2 Technical Platform....9
3. Functional Requirements....10
3.1 Overview....10
3.1.1 User Authentication and Role Separation....14
3.1.2 Knowledge Base Management....14
3.1.3 AI-Assisted Job Creation....15
3.1.4 Job Publishing and CV Collection....16
3.1.5 CV Screening and Requirement-Based Candidate Ranking (F1)....17
3.1.6 AI-Based Candidate Scoring (F2)....18
3.1.7 Shortlisting and Interview Scheduling....19
3.1.8 AI-Powered Video Interview (F3)....19
3.1.9 Real-Time Speech Recognition and Transcription (F4)....20
3.1.10 Candidate Identification and Real-Time Profile Overview (F7)....21
3.1.11 AI-Based Interview Question and Answer Analysis (F5)....22
3.1.12 Consolidated Candidate Assessment (F6)....23
3.1.13 CV Claim Verification Through Interview Analysis (F8)....24
3.1.14 Comprehensive Candidate Interview Scoring (F9)....24
3.1.15 Interview Quality Assessment (F10)....25
3.1.16 Hiring Decision and Candidate Feedback Composer (F11)....26
3.1.17 Candidate Data Deletion and AI Usage Reporting....27
4. User Interface (Screens)....29
4.1 Key Screens....30
5. Non-Functional Requirements....35
5.1 Performance Requirements....35
5.2 Safety Requirements....36
5.3 Security Requirements....37
6. Design Constraints....39
7. Software Quality Attributes....40
8. User Interface (General Guidelines)....41
9. Other Requirements....42
9.1 Requirements Traceability....43

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of HireLucent, an AI-powered recruitment and interview intelligence platform developed as the final evaluation project of the BJIT Academy AI Trainee Developer Programme. It states what the system must do, the quality levels it must meet and the constraints under which it is built, operated and evaluated.

The document is the single, agreed reference for requirement understanding between the trainee developer, the trainer and project manager, and the evaluation panel. It translates the Business Information Document (BID), the approved project clarifications and the client change requests into specific, verifiable requirements. Every requirement carries a unique identifier and an acceptance criterion so that it can be traced to its source, implemented, tested and demonstrated.

This version (1.0) describes the system as delivered for the final submission on 27 October 2026, including all six Core features (F1 to F6), all four Stretch features (F7 to F10) and the Innovation feature (F11).

1.2 Scope

HireLucent supports the hiring workflow end to end, from the creation of a job description to the final, human hiring decision. The platform operates across three stages, pre-interview, during the interview and post-interview, and keeps a human in control of every consequential step. Its core principle is: AI assists, people decide.

Included in scope

- **Job creation:** an AI Agent drafts the job description, the requirement checklist and the scoring weights; the interviewer approves them before the job is saved.
- **Job publishing and CV collection:** the approved job post is published to a Blogger careers page and applications (CV, photo and consent) are collected through a Google Form.
- **Pre-interview:** CV parsing (with OCR for scanned PDFs), semantic requirement matching, must-have screening, multi-dimensional scoring, ATS-style matching and ranking (F1, F2), shortlisting and interview scheduling with calendar invitations.
- **During the interview:** a browser-based one-to-one video interview (F3), live speaker-attributed transcription (F4), candidate identification with interview PIN and face check (F7), suggested questions and a read-only AI Agent in the room.
- **Post-interview:** question and answer analysis (F5), a consolidated, traceable assessment with human review, export, ranking and comparison (F6), CV claim verification (F8), interview performance scoring (F9) and interview quality assessment (F10).
- **Decision and feedback:** hire or reject decisions and personal, fairness-checked feedback emails sent only after approval (F11, Innovation feature).
- **Operations:** single-command Docker Compose startup, candidate data deletion, structured logging and an AI usage and cost report.

Excluded from scope

- Cloud hosting, CI/CD pipelines, load balancing and infrastructure as code (local Docker execution only).

- Enterprise authentication (SSO, LDAP, external identity providers) and any role other than Interviewer and Candidate.
- Integration with commercial applicant tracking (ATS) or HR information (HRIS) systems.
- Offers, payroll and onboarding; the workflow ends at the hiring decision and the candidate feedback email.
- Multi-tenancy, subscription management and billing.
- Native mobile applications; the product is web-based only.
- Training or fine-tuning foundation models; existing AI services are used through their APIs.
- Languages other than English for CVs, interviews and transcripts.

Captured elsewhere

Detailed design (database schema, API contracts, LangGraph node logic and screen layouts) is captured in the README, the architecture and LangGraph diagrams and the OpenAPI documentation listed in Section 1.4, and is referenced rather than repeated here.

1.3 Intended Stakeholder

This document is intended for the following stakeholders:

Stakeholder	Use of this document
Project Manager / Trainer	Approves the requirements, tracks delivery against the WBS and confirms that the delivered system meets the BID and the agreed clarifications.
Evaluation Panel	Uses the requirement identifiers, acceptance criteria and traceability matrix to assess completeness and quality during the final presentation.
Developer (Trainee)	Implements, tests and demonstrates each requirement; keeps the code and this document consistent.
Quality Assurance / Tester	Derives test cases from the acceptance criteria; the test case identifiers in each table link to them.
Interviewer (end user)	Recruiter or hiring manager who uses the platform to create jobs, screen, interview, assess and decide.
Candidate (end user)	Applicant who applies through the form and joins the interview through a private link; has no account and no access to evaluation data.

1.4 References

The documents below were used to prepare this specification or give further detail. Diagrams are held in the docs folder of the GitHub repository.

Reference	Location
Requirement Specification (this document)	SWD/HL/SRS/001, version 1.0
Business Information Document (BID), v1.0	BJIT Academy, AI Trainee Developer Programme, Final Evaluation Assignment (project files)
Project Clarification Log	project clarification.xlsx (trainer answers, project files)
Work Breakdown Structure (WBS)	WBS.xlsx (project files) and Redmine project tree

Reference	Location
README (setup, configuration, privacy, fairness, limitations)	GitHub: Farhan-014/HireLucent, README.md
High-Level Architecture diagram	docs/High_Level_Architecture_30221_Farhan.drawio.pdf
Detailed Architecture diagram	docs/Detailed_Architecture_30221_Farhan.drawio.pdf
LangGraph 1 to 6 workflow diagrams	docs/LangGraph_1_Recruiter_Agent … LangGraph_6_Interview_Quality (PDF)
Innovation feature rationale and design (F11)	docs/innovation_rationale.md, docs/feedback_composer_design.md
API specification (OpenAPI)	http://localhost:8000/docs (running backend)
IEEE 830 / ISO/IEC/IEEE 29148	Recommended practice for software requirements specifications (structure followed)

1.5 Definitions, Acronyms, and Abbreviations

All non-standard terms, acronyms and abbreviations used in this document are defined below.

Term / Acronym	Definition
AI Agent	The conversational assistant (LangGraph 1, Recruiter Agent) that drafts jobs, publishes posts and schedules interviews after approval.
ATS score	Applicant-tracking-style match score from keyword and semantic similarity between a CV and the job.
BID	Business Information Document: the assignment brief that defines the business need.
CV	Curriculum vitae (résumé) submitted by a candidate.
F1 – F11	Feature numbers defined in the BID: Core (F1–F6), Stretch (F7–F10) and Innovation (F11).
HITL	Human-in-the-loop: an AI action that waits for the interviewer's explicit approval.
JD	Job description.
JWT	JSON Web Token used for interviewer sign-in and the candidate pass.
LangGraph	Framework used to build the six stateful, multi-step AI workflows with retries and checkpoints.
LLM	Large language model (Gemini 3.5 Flash-Lite in this project).
MCP	Model Context Protocol: standard interface to the RAG, Google and Blogger tool servers.
OCR	Optical character recognition (Tesseract) for scanned CVs.
PIN	Eight-digit interview code sent with the invitation and typed by the candidate before joining.
RAG	Retrieval-augmented generation: answers checked against Knowledge Base documents stored in ChromaDB.
STT	Speech-to-text (Deepgram or AssemblyAI streaming).
UC	Use case.
WBS	Work Breakdown Structure.

Requirement identifiers. Every requirement is identified as HL_<area>_<number> (for example HL_F1_004), where HL stands for HireLucent and the area is a feature (F1–F11) or a category: AUTH, KB, JOB, PUB, SCH, OPS, NFR, PERF, SAF, SEC, QA, UI, UIG and OTH. The matching test case uses the prefix TC_.

2. Overall Description

2.1 Overview

2.1.1 Product Perspective

HireLucent is a new, self-contained web application. It is not part of an existing product family. It integrates with external services only through clearly defined interfaces: Gemini (language model), Deepgram or AssemblyAI (speech-to-text), Google Workspace (Forms, Drive, Calendar and Gmail) and Blogger (careers page). All other components, including the video server (LiveKit), the vector database (ChromaDB), the embedding model (Ollama) and the face models, run locally in Docker containers.

Figure 1 shows the end-to-end workflow across the four stages, from signing in and creating the job to the final hiring decision and feedback emails.

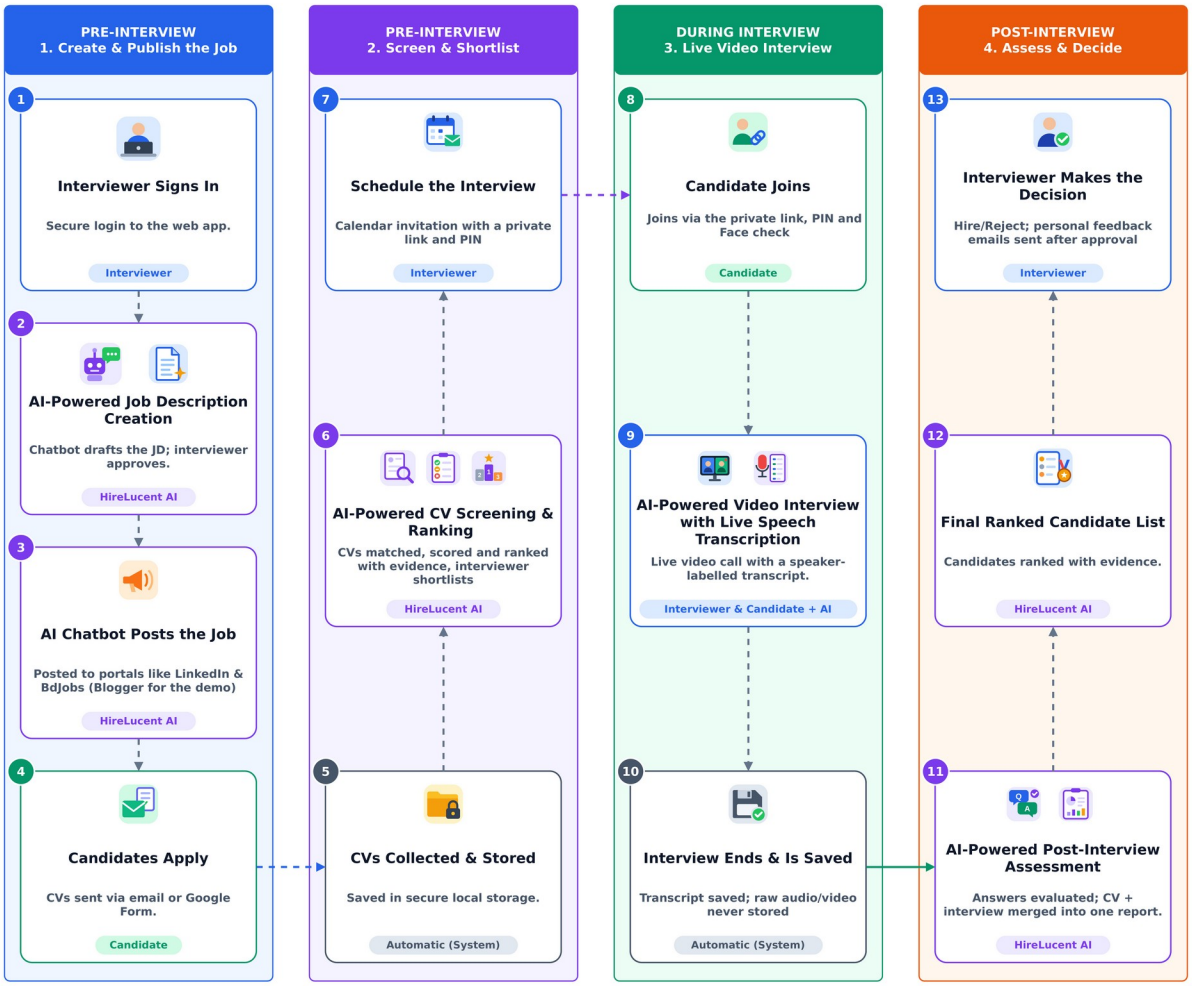


Figure 1: *HireLucent high-level architecture and end-to-end workflow*

2.1.2 Product Functions

At a high level, HireLucent provides the following functions. Section 3 details each of them.

- Create a job with the AI Agent: job description, requirement checklist (must-have and preferred) and scoring weights, approved by the interviewer.
- Publish the job post to a careers page and collect applications through a Google Form.
- Screen, score and rank CVs against the job requirements with evidence; exclude candidates who miss a must-have.
- Shortlist candidates and schedule interviews with calendar invitations, private links and PINs.
- Conduct a one-to-one video interview with live, speaker-attributed transcription and identity verification.
- Analyse every question and answer, write a traceable assessment report and rank interviewed candidates.
- Verify CV claims, score interview performance and assess the quality of the interview itself.
- Record hire or reject decisions and send personal, fairness-checked feedback emails after approval.
- Delete a candidate and all of their data on request, and report AI usage and cost.

2.1.3 User Characteristics

User class	Description	Expertise	Main activities
Interviewer	Recruiter or hiring manager; signed-in user with full access	Comfortable with web applications; no technical or AI knowledge required	Creates and publishes jobs, screens and shortlists, interviews, reviews AI results, approves every external action and makes the final decision
Candidate	Job applicant; no account	General computer user with a browser, camera and microphone	Applies through the form, joins the interview with a private link and PIN, gives consent and takes part in the interview; never sees scores, rankings or reports

2.1.4 Constraints

- Mandatory technology stack: Python, FastAPI with Pydantic, LangGraph and Docker Compose; GitHub with an incremental commit history.
- Local execution only, started with one documented command; no cloud deployment.
- Only the 20 provided sample CVs or synthetic data may be used; no real personal data in the repository, logs or demonstration material.
- Two roles only: Interviewer and Candidate (clarification #18).
- English only for CVs, interviews and transcripts (clarifications #29 and #44).
- Free-tier AI service limits (requests per minute) bound processing speed.
- Solo delivery within the fixed programme schedule; submission on 27 October 2026.

2.1.5 Assumptions and Dependencies

- The interviewer has a Google account with Forms, Drive, Calendar, Gmail and Blogger set up as described in the README; the Google sign-in (Testing mode) is renewed about every seven days.
- Valid API keys for Gemini and Deepgram (or AssemblyAI) are supplied in the .env file.
- The host PC runs Docker Desktop with at least 8 GB of RAM, a camera and a microphone; interviews are demonstrated on one PC with two browser windows (clarification #14).

- A single live interview at a time is sufficient, alongside background processing (BID assumption A9).
- Imperfect AI output is acceptable provided the system stays stable, shows its evidence and keeps the human in control (BID assumption A8).

2.2 Technical Platform

HireLucent runs as a set of Docker containers on the developer's PC. The table lists the platform, frameworks and services used.

Layer	Technology
Frontend	React 19, Vite, Tailwind CSS 4, React Router, livekit-client, MediaPipe Face Detector; served by nginx
Backend API	Python 3.11, FastAPI, Pydantic v2 (OpenAPI documentation), SQLAlchemy 2, Alembic migrations
AI orchestration	LangGraph with PostgreSQL checkpoints: six workflows (Recruiter Agent, CV Screening, Q&A Analysis, Final Assessment, Candidate Feedback, Interview Quality)
Language model	Google Gemini 3.5 Flash-Lite (OpenAI selectable by configuration)
Embeddings and vector store	qwen3-embedding:0.6b through Ollama; ChromaDB 1.5 (cosine similarity)
Tool servers (MCP)	mcp-rag (custom), mcp-google (custom: Gmail, Calendar), mcp-blogger (community server)
Video and audio	Self-hosted LiveKit server (WebRTC); transcription worker joining each room as a hidden participant
Speech-to-text	Deepgram nova-3 (default) or AssemblyAI Universal Streaming, switchable by configuration
Document processing	PyMuPDF, python-docx, Tesseract OCR; PyMuPDF for the report PDF
Face verification	OpenCV YuNet (detection) and SFace (matching), run locally on the CPU
Database	PostgreSQL 16 with named Docker volumes
Deployment	Docker Compose (single command: docker compose up -d --build) on Windows 10/11 with Docker Desktop and WSL 2
Development tools	Visual Studio Code, Git and GitHub, pytest, ESLint, draw.io, Redmine

3. Functional Requirements

3.1 Overview

The table below sums up the main functions provided by HireLucent. Each is detailed in the subsections that follow (3.1.1 to 3.1.17), with its actors and its requirements. Three UML use case diagrams, one for each stage of the hiring workflow, show the actors, the main use cases and how they relate (Figures 2 to 4).

Serial No	Main Features	Description
1	User Authentication and Role Separation	Interviewer sign-in and strict separation of interviewer and candidate capabilities.
2	Knowledge Base Management	Upload of job description templates, reference documents and question banks used for retrieval.
3	AI-Assisted Job Creation	AI Agent drafts the job description, requirements and weights; the interviewer approves.
4	Job Publishing and CV Collection	Approved post published to the careers page; applications collected from the Google Form.
5	CV Screening and Ranking (F1)	Semantic requirement matching with evidence, must-have exclusion and ranking.
6	AI-Based Candidate Scoring (F2)	Weighted 0–100 scores per dimension with justification, combined with an ATS score.
7	Shortlisting and Interview Scheduling	Human shortlist and future-dated interviews with calendar invitation, link and PIN.
8	AI-Powered Video Interview (F3)	Browser-based one-to-one video session with full lifecycle and in-call controls.
9	Live Speech Recognition and Transcription (F4)	Speaker-attributed, timestamped transcript produced and saved live.
10	Candidate Identification and Profile (F7)	Interview PIN, consent and face check, with a profile panel for the interviewer.
11	Interview Q&A Analysis (F5)	Question–answer segmentation, evaluation with evidence and requirement coverage.
12	Consolidated Candidate Assessment (F6)	Traceable report with human review, ranking, export and comparison.
13	CV Claim Verification (F8)	Each CV claim compared with what the candidate said in the interview.
14	Comprehensive Interview Scoring (F9)	Interview performance score across six evidence-based dimensions.
15	Interview Quality Assessment (F10)	Quality of the interview process, recommendations and cross-candidate consistency.
16	Decision and Feedback Composer (F11)	Hire or reject decisions and personal, fairness-checked feedback emails after approval.
17	Data Deletion and AI Usage Reporting	Complete deletion of a candidate’s data and a report of AI calls and cost.

Use case diagrams. The diagrams follow UML 2.5 conventions: stick figures are human actors, boxes marked «system» are external systems acting as secondary actors, ellipses are use cases inside the HireLucent system boundary, solid lines are associations, and dashed arrows are «include» (always performed) or «extend» (performed only in some cases) dependencies. Feature numbers F1–F11 link each use case to the BID.

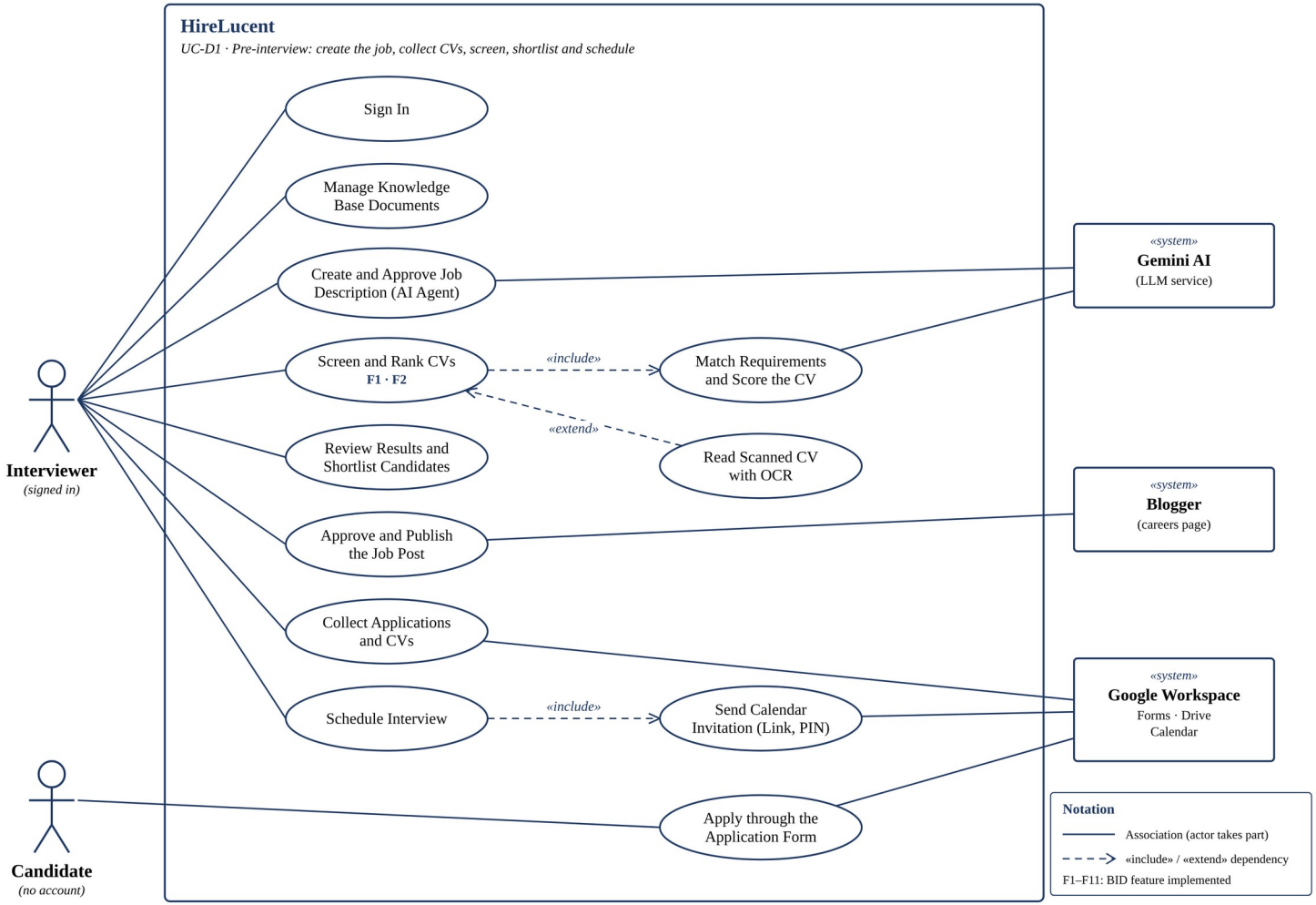


Figure 2: *UC-D1, pre-interview use cases: create the job, collect CVs, screen, shortlist and schedule*

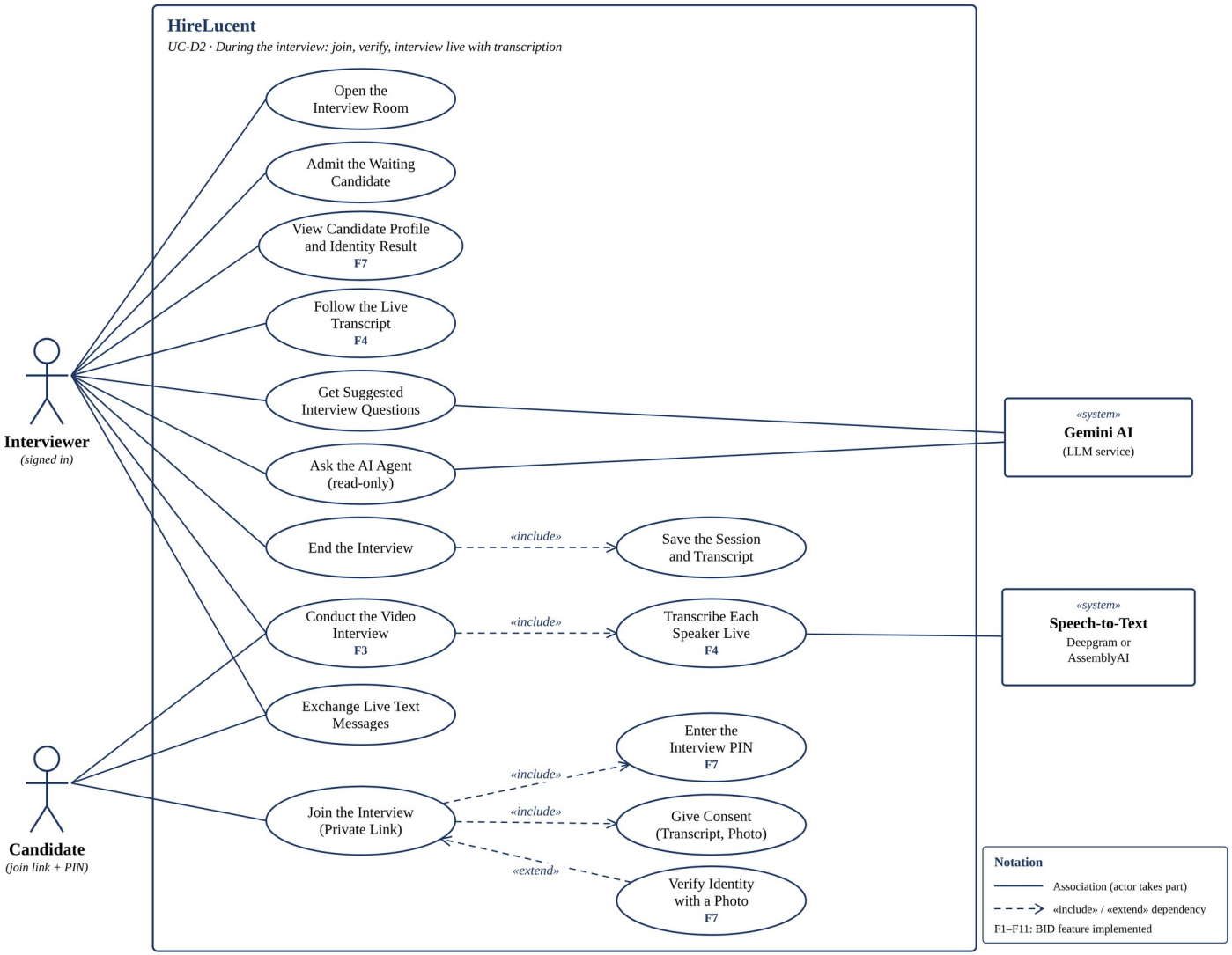


Figure 3: *UC-D2, during-interview use cases: join, verify identity, interview live with transcription*

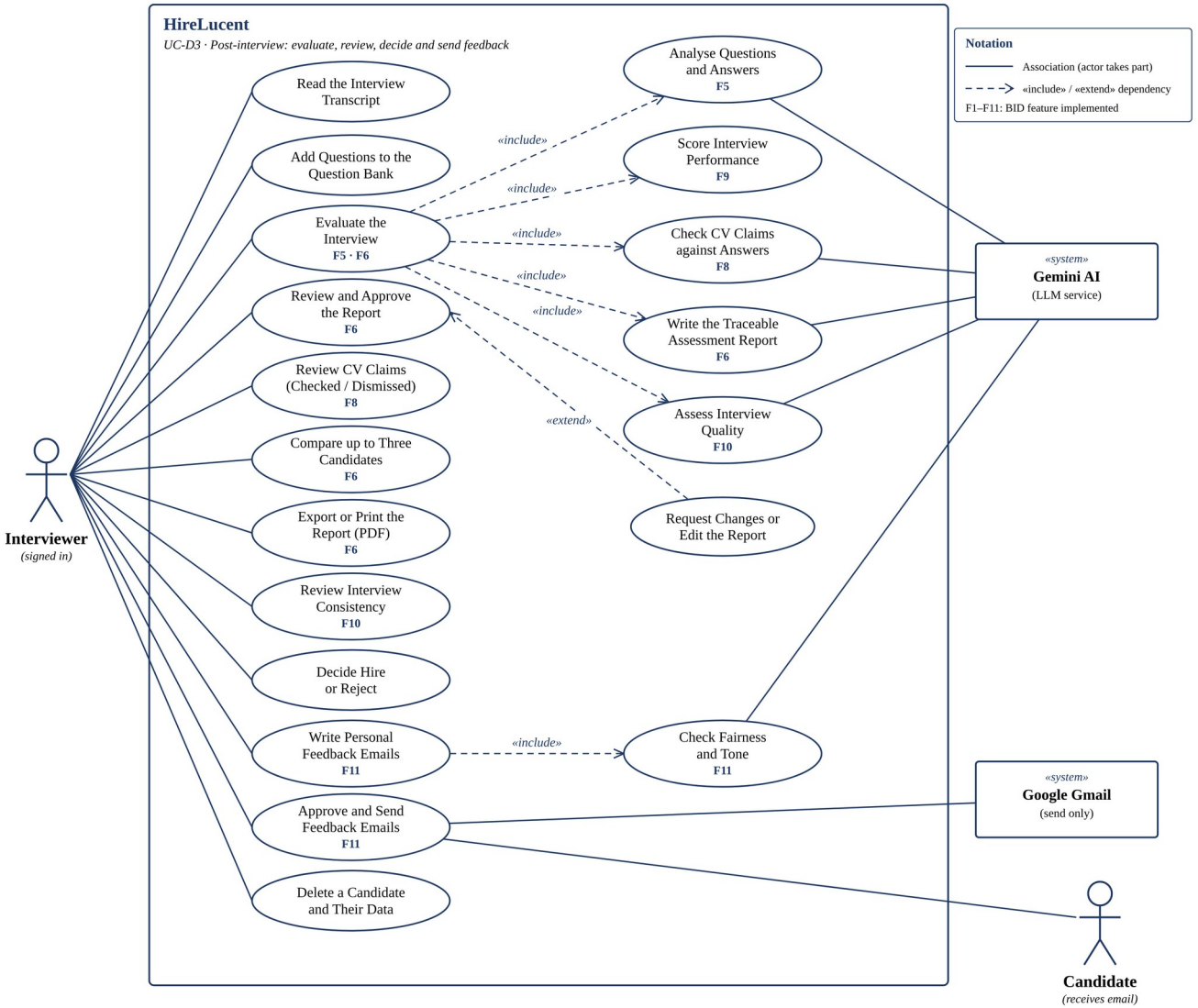


Figure 4: *UC-D3, post-interview use cases: evaluate, review, decide and send feedback*

3.1.1 User Authentication and Role Separation

This function controls who can use HireLucent and what each role can reach. Actors: Interviewer (signed in) and Candidate (no account, interview access only). It supports BID requirement N9 and clarifications #5 and #18.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_AUTH_001	The system shall let an interviewer sign in with an email address and password and issue a signed access token (JWT).	Valid credentials open the dashboard; invalid credentials show a clear error and issue no token.	Token valid for 60 minutes; no self-registration; the first interviewer is created from the .env settings.	TC_AUTH_001
HL_AUTH_002	The system shall store interviewer passwords only as Argon2 hashes.	No plain-text password exists in the database, logs or responses.	Password reset by email is out of scope.	TC_AUTH_002
HL_AUTH_003	The system shall support exactly two roles: Interviewer (full access) and Candidate (interview only).	Every API route belongs to one role or is explicitly public.	No administrator or recruiter role (clarification #18).	TC_AUTH_003
HL_AUTH_004	The system shall refuse every scoring, ranking, assessment and transcript request made with a candidate pass.	A candidate pass receives HTTP 403 on all interviewer routes; requests without a token receive 401.	Verified automatically for every route by the access-separation test.	TC_AUTH_004
HL_AUTH_005	The system shall give a candidate access to one interview only, through a private link, a candidate pass and a room token.	A pass for one interview cannot open another interview or any interviewer page.	Links stop working when the interview is rescheduled, cancelled or ended.	TC_AUTH_005

3.1.2 Knowledge Base Management

The Knowledge Base holds the reference material that the AI uses through retrieval (RAG): job description templates for style, authoritative reference documents for checking answer accuracy (clarification #10), and question banks for the interview room. Actor: Interviewer.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_KB_001	The system shall let the interviewer upload a document with a type: job description template, reference document or question bank.	The document appears in the list with its type and processing status.	PDF, DOCX, TXT or MD; at most 10 MB per file.	TC_KB_001

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_KB_002	The system shall extract, chunk, embed and store each uploaded document in the vector collection of its type, in the background.	Status moves from processing to ready; a failure shows a readable reason.	Collections are fixed in code and never chosen by the AI.	TC_KB_002
HL_KB_003	The system shall allow roles to be set on question bank files so that questions are offered only for matching job titles.	An empty role list makes the file general; a role matches when it is part of the job title.	Matching is by title words, not by meaning.	TC_KB_003
HL_KB_004	The system shall delete a document together with all of its stored vectors.	After deletion the document is no longer used by any search.	—	TC_KB_004

3.1.3 AI-Assisted Job Creation

The interviewer creates a job by talking to the AI Agent (LangGraph 1), which drafts the job description using similar templates from the Knowledge Base and proposes a structured requirement profile. The job is saved only after the interviewer approves it. Actors: Interviewer, Gemini AI. Supports BID Section 4.1 (job requirement definition) and clarification #35.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_JOB_001	The system shall provide an AI Agent chat that streams its replies and keeps one conversation per job.	Replies appear progressively; earlier conversations can be reopened.	English only.	TC_JOB_001
HL_JOB_002	The AI Agent shall draft a job description with the sections About the Role, Responsibilities, Requirements and Nice to Have, using matching job description templates as style references.	The draft is shown with its version and the templates used; the interviewer can request changes or edit it directly.	Quality depends on the AI service; every draft is reviewed by the interviewer.	TC_JOB_002
HL_JOB_003	The system shall build a requirement profile: minimum years of experience and requirements R1, R2 … with category, priority (must-have or preferred) and keywords.	The profile is shown in a confirmation card before saving.	At most 40 requirements and 25 keywords per requirement; experience in whole years.	TC_JOB_003

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_JOB_004	The AI Agent shall propose scoring weights for technical skills, experience and qualifications that total 100%.	The interviewer can accept or change the weights before saving (clarification #22).	Weights apply to the job as a whole.	TC_JOB_004
HL_JOB_005	The system shall save the job only after the interviewer approves the save action, and give it a unique job code.	Nothing is saved without approval; the job brief shows the job code.	Human-in-the-loop action.	TC_JOB_005
HL_JOB_006	The system shall let the interviewer edit the requirements and weights after saving, and reset the screening when they change.	Changed requirements trigger a fresh screening.	Editing is locked once the job has interviews.	TC_JOB_006
HL_JOB_007	The system shall support several jobs at the same time, each with its own candidates and results.	The Jobs page lists all jobs with their status.	Clarification #41.	TC_JOB_007

3.1.4 Job Publishing and CV Collection

Per the client change request, the AI Agent publishes the approved job post to a careers page with an API (Blogger, standing in for job portals such as LinkedIn and BdJobs) and candidates apply through a Google Form. Actors: Interviewer, Candidate, Blogger, Google Workspace.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_PUB_001	The system shall check the job's Google Form before publishing: it must be readable and contain required questions for full name, email, CV upload, photo upload and consent.	Publishing is refused with the missing question named.	One form per job; questions are found by a word in their title.	TC_PUB_001
HL_PUB_002	The system shall show a preview of the job post and publish it to the careers page only after the interviewer approves.	After approval the post is live and its link is saved with the job.	Blogger is a demonstration stand-in for real job portals.	TC_PUB_002
HL_PUB_003	The system shall fetch new applications from the form on request, download each CV and create a candidate record in the background.	New candidates appear on the Applications tab with progress feedback.	Requires a valid Google sign-in.	TC_PUB_003

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_PUB_004	The system shall skip duplicate applications and applicants who were deleted.	Fetching twice does not create a second record for the same applicant.	—	TC_PUB_004
HL_PUB_005	With the candidate's consent, the system shall check the application photo and store a face code for the identity check (F7).	A photo problem is recorded with its reason but never stops the application.	Biometric data handled as described in HL_SEC_009.	TC_PUB_005

3.1.5 CV Screening and Requirement-Based Candidate Ranking (F1)

The CV Screening workflow (LangGraph 2) reads every CV, extracts structured information and matches it semantically against the job's requirements, producing evidence for each requirement and a ranking. Actors: Interviewer, Gemini AI. Supports BID F1 and clarifications #6, #9, #19 and #37.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F1_001	The system shall screen a batch of CVs as a background task with visible progress.	All CVs of the job are processed; progress is shown until the task finishes.	At most 4 CVs in parallel; pace limited by the AI service.	TC_F1_001
HL_F1_002	The system shall extract text from PDF and DOCX CVs and use OCR for scanned PDFs.	A scanned PDF with readable text is parsed; an unreadable CV is recorded as failed without stopping the batch.	English CVs only.	TC_F1_002
HL_F1_003	The system shall extract structured CV information: contact details, education, work history, skills, certifications and projects.	Parsed facts are shown for each candidate.	Contacts are masked before any text is sent to the AI.	TC_F1_003
HL_F1_004	The system shall mark each requirement as met, partially met or not met using semantic understanding, with an explanation and supporting evidence from the CV.	A CV stating related experience in other words is matched to the requirement.	AI-generated explanation is sufficient (clarification #37).	TC_F1_004
HL_F1_005	The system shall exclude a candidate who does not meet a must-have requirement, and record the reason.	Excluded candidates are listed separately with the missing must-have.	Keywords first; the AI is asked only when no keyword is found, so no one is excluded on a missing keyword alone.	TC_F1_005

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F1_006	The system shall rank the screened candidates by requirement match and by final score.	Two views: Requirement match and Final ranking, with search and expandable rows.	The ranking never shortlists anyone automatically.	TC_F1_006
HL_F1_007	A screening interrupted by a restart shall resume each CV from the step it had reached.	No CV is parsed or matched twice after a restart.	LangGraph checkpoints in PostgreSQL.	TC_F1_007

3.1.6 AI-Based Candidate Scoring (F2)

Each screened candidate receives an objective, explainable suitability score. Actors: Interviewer, Gemini AI. Supports BID F2 and clarifications #11, #20 and #22.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F2_001	The system shall score technical skills, relevant experience and qualifications on a 0–100 scale, each with a short justification.	Every dimension shows its score and justification.	Scale fixed at 0–100 (clarification #20).	TC_F2_001
HL_F2_002	The system shall combine the dimension scores using the job's weights into a criteria score.	Changing the weights changes the criteria score accordingly.	Weights total 100%.	TC_F2_002
HL_F2_003	The system shall compute an ATS-style match score from keyword and semantic similarity.	The ATS score is shown beside the criteria score.	Clarification #11.	TC_F2_003
HL_F2_004	The system shall compute the final score as 0.8 × criteria score + 0.2 × ATS score and show the full breakdown.	The breakdown adds up to the displayed final score.	—	TC_F2_004
HL_F2_005	Scores shall be stable on repeated runs for the same CV and job.	Repeated scoring differs by no more than about 2 points.	The language model ignores temperature; documented as a known limitation.	TC_F2_005
HL_F2_006	Scores shall be saved and reused by the consolidated assessment.	Scores survive a restart and appear in the report.	—	TC_F2_006

3.1.7 Shortlisting and Interview Scheduling

The interviewer decides who is interviewed and schedules future-dated interviews, either from the job's Interviews tab or through the AI Agent with approval cards. Actors: Interviewer, Google Workspace (Calendar). Supports clarifications #39 and #40.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_SCH_001	The system shall let the interviewer shortlist candidates one by one, by top N or by minimum score, and clear the shortlist.	Only shortlisted candidates can be scheduled.	No automatic shortlisting.	TC_SCH_001
HL_SCH_002	The system shall schedule an interview for a future date and time with a length of 30, 45 or 60 minutes.	The interview appears under Upcoming with its local time.	Minimum notice 15 minutes; no overlapping interviews for the same interviewer.	TC_SCH_002
HL_SCH_003	The system shall send a calendar invitation containing a private join link and an eight-digit interview PIN, with a message the interviewer can edit and preview.	The candidate receives the invitation; the preview hides the real link and PIN.	Only hashes of the link token and PIN are stored.	TC_SCH_003
HL_SCH_004	The system shall let the interviewer reschedule or cancel an interview, sending an updated invitation or a cancellation.	Rescheduling issues a new link and PIN; the old link stops working.	Confirmation dialog before each change.	TC_SCH_004
HL_SCH_005	Scheduling, rescheduling and cancelling through the AI Agent shall wait for the interviewer's approval.	No calendar event is created without approval.	Human-in-the-loop action.	TC_SCH_005

3.1.8 AI-Powered Video Interview (F3)

Interviews take place in the browser on a self-hosted LiveKit video server. The interviewer room combines the video, the candidate profile, the live transcript and the AI Agent. Actors: Interviewer, Candidate. Supports BID F3 and clarifications #3, #26 and #34.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F3_001	The system shall provide a browser-based one-to-one video and audio session between the interviewer and the candidate.	Both participants see and hear each other in the room.	One interviewer and one candidate per session.	TC_F3_001

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F3_002	The system shall support the session lifecycle: schedule, join, wait, admit, conduct and end.	The candidate waits until admitted; ending closes the room and stops all candidate access.	Join opens 10 minutes before the start.	TC_F3_002
HL_F3_003	The system shall provide in-call controls: mute, camera on/off, screen share, full screen and end.	Each control works for its participant.	Only the interviewer can end the interview.	TC_F3_003
HL_F3_004	The system shall reconnect automatically after a network interruption without losing the transcript captured so far.	After a drop the session resumes and saved transcript lines remain.	A "still trying" message appears after 30 seconds.	TC_F3_004
HL_F3_005	The system shall associate every session with one candidate and one job and persist start and end times and participants.	Session details are shown on the Interviews tab and in the report.	—	TC_F3_005
HL_F3_006	The system shall offer live text messages between interviewer and candidate, and a read-only AI Agent for the interviewer.	Messages reach the other participant; the AI Agent cannot perform external actions in the room.	Live messages are not stored.	TC_F3_006
HL_F3_007	The system shall generate suggested interview questions on request from the candidate’s CV and screening results.	Five questions per request; each can be marked as asked.	Generated only when requested.	TC_F3_007

3.1.9 Real-Time Speech Recognition and Transcription (F4)

A transcription worker joins each live interview as a hidden participant and streams the interviewer’s and the candidate’s microphones separately to the speech-to-text service, so each line is attributed to the right speaker without diarisation. Actors: Interviewer, Speech-to-Text service. Supports BID F4 and clarifications #4, #8, #31 and #36.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F4_001	The system shall transcribe both participants during the session with minimal perceptible lag.	Words appear while being spoken; saved lines arrive within 2 seconds of the speaker stopping.	English only; depends on the speech-to-text service.	TC_F4_001
HL_F4_002	The system shall attribute every utterance to the interviewer or the candidate from their separate audio channel.	Each transcript line shows the correct speaker.	Clarification #4.	TC_F4_002

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F4_003	The system shall timestamp every turn from the start of the interview and keep the clock time.	Each line shows its elapsed time; hovering shows the clock time.	Clarification #31.	TC_F4_003
HL_F4_004	The system shall save each turn immediately and keep overlapping speech as separate entries.	No saved line is lost if the session or the worker restarts.	—	TC_F4_004
HL_F4_005	The system shall show a live transcript panel to the interviewer only, with listening, paused and not-available states.	The candidate never receives transcript data.	Accepted only from the hidden worker, so the candidate cannot fake lines.	TC_F4_005
HL_F4_006	The system shall provide a read-only transcript viewer with search, timeline and text download after the interview.	Search highlights matches; the download contains timestamped speaker lines.	No audio playback; file name contains no candidate name.	TC_F4_006
HL_F4_007	The system shall support two switchable speech-to-text providers.	Changing the setting and restarting the worker switches provider.	Deepgram nova-3 (default) or AssemblyAI.	TC_F4_007

3.1.10 Candidate Identification and Real-Time Profile Overview (F7)

The candidate proves their identity with the interview PIN and a face check against their application photo, and the interviewer sees a compact profile panel in the room. The trainer approved going beyond clarification #21. Actors: Candidate, Interviewer. Supports BID F7 and N17.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F7_001	The system shall ask the candidate for the eight-digit interview PIN before joining.	A correct PIN continues; five wrong attempts lock the link for 15 minutes.	Only the PIN hash is stored.	TC_F7_001
HL_F7_002	The system shall ask for explicit consent to the transcript and the identity photo before joining.	The candidate cannot continue without consent.	Consent is also collected in the application form.	TC_F7_002
HL_F7_003	The system shall guide the candidate to take a photo with live tips and compare it with the application photo and the job’s other applicants.	Result shown to the interviewer as verified, low or very low match, another applicant, or not checked.	The photo taken when joining is never stored; three tries, then the candidate continues.	TC_F7_003
HL_F7_004	The identity check shall never block the candidate from joining.	Low-confidence or failed checks are shown to the interviewer as advice only.	No liveness check (known limitation).	TC_F7_004

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F7_005	The system shall show the interviewer a profile panel: application photo, identity result, key details, pre-interview score and areas to probe.	Areas to probe list requirements partly met or not met.	Interviewer only.	TC_F7_005

3.1.11 AI-Based Interview Question and Answer Analysis (F5)

After the interview the Q&A Analysis workflow (LangGraph 3) segments the transcript into question–answer pairs and evaluates each job-related answer with evidence, checking technical and company answers against reference documents. Actors: Interviewer, Gemini AI. Supports BID F5 and clarifications #10, #16 and #38.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F5_001	The system shall segment the transcript into question–answer pairs, including follow-ups and multi-part questions, and classify each as technical, behavioural, company or small talk.	Each pair links to its transcript turns; follow-ups are indented under their question.	A question lost from the transcript is not evaluated (known limitation).	TC_F5_001
HL_F5_002	The system shall evaluate each job-related answer for relevance, accuracy, completeness and articulation on a 0–100 scale.	Every answer shows four criteria and an answer score.	Small talk is transcribed but not scored (clarification #16).	TC_F5_002
HL_F5_003	The system shall check technical and company answers against Knowledge Base reference documents and show the accuracy source.	The source shows reference, AI judgement or not checked.	Company accuracy is never guessed without a reference.	TC_F5_003
HL_F5_004	Each evaluation shall include a written justification and an exact evidence quote from the transcript.	A quote not found in the answer is rejected and retried.	Up to three attempts; untrusted parts are dropped, not kept.	TC_F5_004
HL_F5_005	The system shall map answers to the job requirements and show requirements covered, weakly covered and never explored.	Requirement coverage tab lists every requirement with its status.	—	TC_F5_005

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F5_006	The system shall suggest the interviewer's own good questions for the question bank, added only when the interviewer picks them.	Nothing is added to the bank automatically.	—	TC_F5_006

3.1.12 Consolidated Candidate Assessment (F6)

The Final Assessment workflow (LangGraph 4) merges screening, scoring and interview analysis into one traceable report that the interviewer reviews before it counts. Actors: Interviewer, Gemini AI. Supports BID F6 and clarification #49.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F6_001	The system shall compute an overall score as 0.4 × CV score + 0.6 × interview score and a label: strong fit (≥ 70), possible fit (≥ 60) or not a fit.	Component scores are always shown beside the overall score.	The label is decision support only.	TC_F6_001
HL_F6_002	The system shall write a summary, strengths and weaknesses, evidence gaps and areas to verify.	The report shows all sections.	At most five strengths and five weaknesses.	TC_F6_002
HL_F6_003	Every strength and weakness shall cite a transcript quote or a CV requirement result; untraceable statements shall be removed.	Each point shows its source chips.	Checked by code after the AI writes the report.	TC_F6_003
HL_F6_004	The report shall stay a draft until the interviewer approves it, requests changes with a note, or edits it.	Approval records who approved and when.	Edits may change the summary and remove points only.	TC_F6_004
HL_F6_005	The system shall rank interviewed candidates by overall score with filters and counts.	Ranked candidates view lists drafts and approved reports.	—	TC_F6_005
HL_F6_006	The system shall export the report as a PDF and support printing.	The PDF contains all report sections; the file name has no candidate name.	—	TC_F6_006
HL_F6_007	The system shall compare up to three ranked candidates side by side with the best values highlighted.	Scores, requirements and interview dimensions appear in one view.	Maximum three candidates.	TC_F6_007

3.1.13 CV Claim Verification Through Interview Analysis (F8)

Every claim in the CV is compared with what the candidate said in the interview, as part of the Final Assessment workflow. Results are prompts for human follow-up, never accusations. Actors: Interviewer, Gemini AI. Supports BID F8.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F8_001	The system shall extract every claim from the parsed CV: total experience, each job, project, certification, education entry and skill.	Claims are listed with identifiers C1, C2 …	—	TC_F8_001
HL_F8_002	The system shall classify each claim as contradiction, unverified, not discussed, consistent or could not be checked.	Each result shows the CV text, the question asked, the candidate's words and a neutral reason.	Accusing wording is refused by code.	TC_F8_002
HL_F8_003	Each quote shall be verified against the answer it came from, with a link to that moment in the transcript.	Opening the link shows the highlighted transcript turn.	Up to three attempts, then not checked.	TC_F8_003
HL_F8_004	The interviewer shall be able to mark a contradiction or unverified claim as checked or dismissed with an optional note.	The review shows who and when, with Undo.	Note up to 500 characters.	TC_F8_004
HL_F8_005	Claim results shall never change scores, the report or the decision; approving with an open contradiction shows a warning only.	Scores are identical with and without claim results.	—	TC_F8_005

3.1.14 Comprehensive Candidate Interview Scoring (F9)

The interview performance score is distinct from the CV score and rests on six dimensions computed by code from the evaluated answers and the requirement coverage. Actors: Interviewer, Gemini AI (per-answer judgements). Supports BID F9.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F9_001	The system shall score technical competency, communication, relevance of responses, problem-solving approach, depth of experience and alignment with job requirements on a 0–100 scale.	The Interview performance tab shows six bars with a “based on” line.	Equal weights, fixed in code.	TC_F9_001
HL_F9_002	Each dimension shall show the answers or requirements it rests on, with quotes and links to the transcript.	Opening a dimension lists its evidence.	—	TC_F9_002
HL_F9_003	A dimension without evidence shall be marked “not assessed”, never scored as zero.	Not-assessed dimensions are left out of the interview score.	—	TC_F9_003
HL_F9_004	The interview score shall be the average of the assessed dimensions and require at least three assessed dimensions.	With fewer than three, “not enough interview evidence” is shown and no interview or overall score is given.	—	TC_F9_004
HL_F9_005	Communication shall be judged on clarity and structure only, never on accent or fluency.	Fairness instructions are present in every prompt.	Checked by the fairness test.	TC_F9_005

3.1.15 Interview Quality Assessment (F10)

The Interview Quality workflow (LangGraph 6) evaluates how the interview was run, not the candidate, and recommends improvements for the next interview. Actors: Interviewer, Gemini AI (recommendations only). Supports BID F10.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F10_001	The system shall compute a 0–100 quality score from three equal parts: questions, requirement coverage and interaction, with a label good (≥ 70), needs improvement (≥ 50) or poor.	The Interview quality tab shows the score, label and three parts.	Never changes the candidate’s scores or decision.	TC_F10_001

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F10_002	The system shall flag questions that produced little signal and requirements left unexplored or weakly explored.	Each finding links to the question or requirement.	Little signal: not answered, under 5 words, or relevance below 40.	TC_F10_002
HL_F10_003	The system shall measure speaking time per participant, the longest interviewer stretch, small talk and the interview length against the plan.	A speaking bar and a two-lane timeline are shown; a bar opens the transcript at that turn.	Candidate share below 60% is flagged; above 70% is information only.	TC_F10_003
HL_F10_004	The system shall write up to five actionable recommendations, each traced to its findings, with an example question where useful.	Every number and identifier in a recommendation appears in its findings.	If the AI is unavailable, findings are still shown and the check can be repeated.	TC_F10_004
HL_F10_005	The system shall compare interviews of the same job for consistency and highlight the ones that differ.	The Interview consistency view shows the typical interview and the differing interviews.	Shown from two checked interviews.	TC_F10_005

3.1.16 Hiring Decision and Candidate Feedback Composer (F11)

The Innovation feature: after the decision, every candidate receives a personal, respectful feedback email written from their own job-related evidence, checked by code and by an AI review, and sent only after the interviewer approves it. Actors: Interviewer, Candidate (recipient), Gemini AI, Google Gmail. Supports BID Section 6 (C1–C8).

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F11_001	The system shall record a hire or reject decision once the report is approved, per candidate or by rule (top N or minimum score).	The decision is shown on the Assessments tab and can change until its email is sent.	Hire or Reject only.	TC_F11_001
HL_F11_002	Candidates not shortlisted after screening shall be treated as rejected at screening and receive a feedback email.	They appear on the Feedback emails view of the Screening tab.	Shortlisting is locked once their email is sent.	TC_F11_002

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_F11_003	The system shall draft a personal email for each rejected candidate from their strengths and growth areas, and use fixed messages for hires and unreadable CVs.	Each draft is different and mentions the candidate's own evidence.	The AI never receives the candidate's name, email, scores, other candidates or transcript quotes.	TC_F11_003
HL_F11_004	Every draft shall pass a code fairness check and an AI review of tone and unsupported claims; remaining problems are shown as warnings naming the exact words.	Drafts with warnings are highlighted and are not sent by "Send all passed".	Up to three rewrites before a warning is kept.	TC_F11_004
HL_F11_005	Nothing shall be sent until the interviewer approves, edits, requests changes, discards or confirms "Send anyway".	Approved emails are sent through Gmail; a failed send can be retried.	Human-in-the-loop action.	TC_F11_005

3.1.17 Candidate Data Deletion and AI Usage Reporting

These functions support privacy (BID N19) and cost awareness for the AI services (constraint CN7). Actor: Interviewer.

Requirements

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_OPS_001	The system shall delete a candidate and all of their data: CV file and vectors, photo and face code, screening results, interviews, transcripts, assessments, feedback email and AI workflow state.	After deletion no record or file of the candidate remains.	Refused while the candidate has an upcoming or running interview.	TC_OPS_001
HL_OPS_002	Deleting a job shall delete all of its candidates and their data in the same way.	The job and everything linked to it are removed.	Refused while the job has upcoming interviews.	TC_OPS_002
HL_OPS_003	The system shall record the token counts of every language model answer with its route, workflow step and job, candidate and interview IDs.	One usage line per answer; no prompt or answer text is recorded.	Calls that receive no answer are not counted.	TC_OPS_003

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_OPS_004	The system shall produce an AI cost report as CSV files: per call, and totals by day, route and step, with unit costs per CV, interview, email, job and hire and a monthly projection.	Report totals equal the sum of the recorded calls; reports can be filtered by date or job.	Prices come from configuration and must be checked against the provider’s pricing.	TC_OPS_004

4. User Interface (Screens)

HireLucent offers the following screens. Interviewer screens are reached after sign-in; candidate screens are reached only through the private interview link. Each screen is linked to the functional requirements it serves.

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
HL_UI_001	Landing Page	—	Product introduction with product tour, how-it-works steps, features and principles; entry to sign-in.	TC_UI_001
HL_UI_002	Sign-In	HL_AUTH_001	Interviewer sign-in with email, password and show/hide; note for candidates.	TC_UI_002
HL_UI_003	Dashboard	HL_JOB_007	Overview of the hiring activity and quick access to jobs.	TC_UI_003
HL_UI_004	Jobs	HL_JOB_007	List of all jobs with status and job code.	TC_UI_004
HL_UI_005	AI Agent	HL_JOB_001 – 005, HL_SCH_005	Streaming chat with drafts, approval cards and a live job brief.	TC_UI_005
HL_UI_006	Knowledge Base	HL_KB_001 – 004	Upload, type, roles, status and deletion of reference documents.	TC_UI_006
HL_UI_007	Job – Job Post tab	HL_PUB_001 – 002	Job description, form check and published post link.	TC_UI_007
HL_UI_008	Job – Applications tab	HL_PUB_003 – 004	Fetch new applications and list candidates with progress feedback.	TC_UI_008
HL_UI_009	Job – Screening tab	HL_F1_001 – 006, HL_F2_001 – 004, HL_SCH_001	Final ranking, requirement match and feedback email views; shortlisting.	TC_UI_009
HL_UI_010	Job – Interviews tab	HL_SCH_002 – 004, HL_F5_001	Upcoming, completed and cancelled interviews; schedule, reschedule, cancel, new PIN, evaluate.	TC_UI_010
HL_UI_011	Job – Assessments tab	HL_F6_005, HL_F10_005, HL_F11_001	Ranked candidates, interview consistency and decisions.	TC_UI_011
HL_UI_012	Interviewer Interview Room	HL_F3_001 – 007, HL_F4_005, HL_F7_005	Video, controls, candidate profile, suggested questions, live transcript, AI Agent and messages.	TC_UI_012
HL_UI_013	Transcript Viewer	HL_F4_006	Searchable transcript with timeline and text download.	TC_UI_013
HL_UI_014	Assessment – Report tab	HL_F6_001 – 004, HL_F6_006	Scores, label, summary, strengths, weaknesses, gaps; approve, request changes, edit, export.	TC_UI_014

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
HL_UI_015	Assessment – Interview performance tab	HL_F9_001 – 004	Six dimension bars with evidence.	TC_UI_015
HL_UI_016	Assessment – Q&A analysis tab	HL_F5_001 – 004	Questions, answers, criteria, justification and evidence.	TC_UI_016
HL_UI_017	Assessment – Requirement coverage tab	HL_F5_005 – 006	Coverage per requirement with CV status and question bank suggestions.	TC_UI_017
HL_UI_018	Assessment – CV claims tab	HL_F8_001 – 005	Claim results with evidence and review actions.	TC_UI_018
HL_UI_019	Assessment – Interview quality tab	HL_F10_001 – 004	Quality score, recommendations, interaction and comparison.	TC_UI_019
HL_UI_020	Assessment – Decision tab	HL_F11_001	Hire or reject decision for the approved report.	TC_UI_020
HL_UI_021	Email Review Panel	HL_F11_003 – 005	Review, edit, request changes, discard, approve and send feedback emails.	TC_UI_021
HL_UI_022	Compare Candidates	HL_F6_007	Up to three candidates side by side.	TC_UI_022
HL_UI_023	Candidate – PIN Page	HL_F7_001	Interview PIN entry with lock-out message.	TC_UI_023
HL_UI_024	Candidate – Pre-Join	HL_F7_002, HL_F3_002	Job and time, camera and microphone test, consent.	TC_UI_024
HL_UI_025	Candidate – Identity Photo	HL_F7_003 – 004	Camera oval with live tips and automatic photo.	TC_UI_025
HL_UI_026	Candidate – Waiting Room	HL_F3_002	Waiting until the interviewer admits the candidate.	TC_UI_026
HL_UI_027	Candidate – Interview Room	HL_F3_001, HL_F3_003, HL_F3_006	Video with labelled controls and message panel.	TC_UI_027
HL_UI_028	Candidate – Interview Ended	HL_F3_002	Confirmation that the interview has ended.	TC_UI_028

4.1 Key Screens

The figures below show the key screens of HireLucent. Together they cover every BID feature from F1 to F11, from creating a job to sending feedback. All names and data shown are fictional.

Getting started and job creation

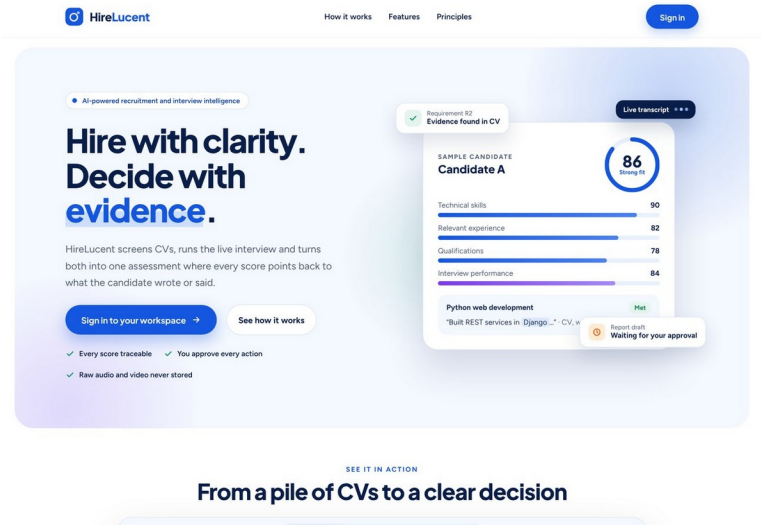


Figure 5: *Landing page (HL_UI_001)*

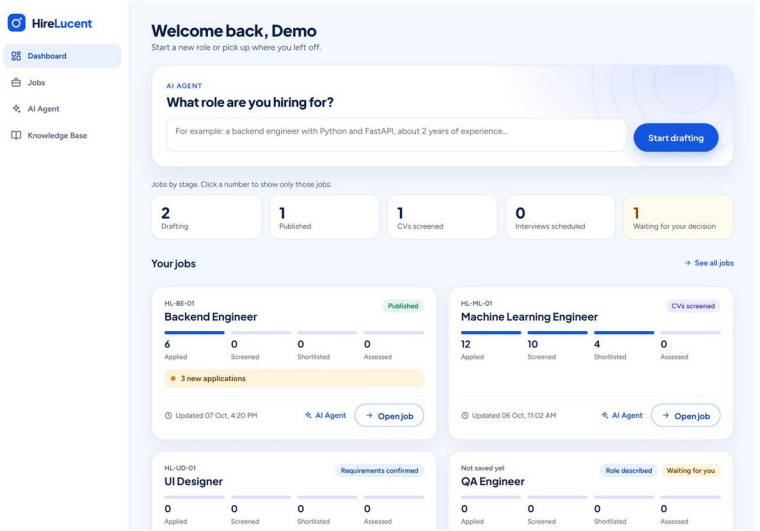


Figure 6: *Dashboard with jobs by stage (HL_UI_003)*

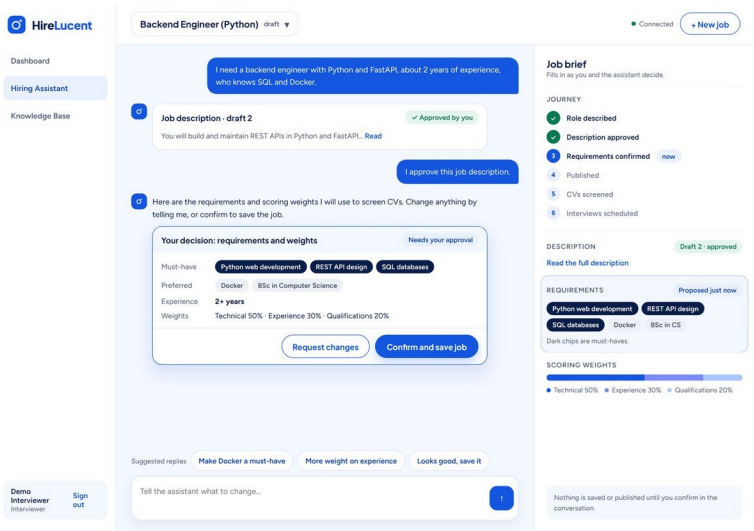


Figure 7: *AI Agent: job description and requirements card (HL_UI_005)*

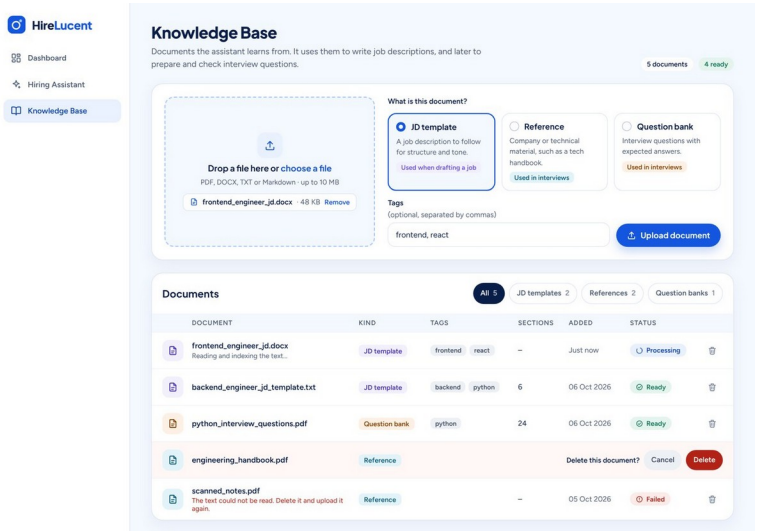


Figure 8: *Knowledge Base (HL_UI_006)*

Screening, scheduling and the live interview

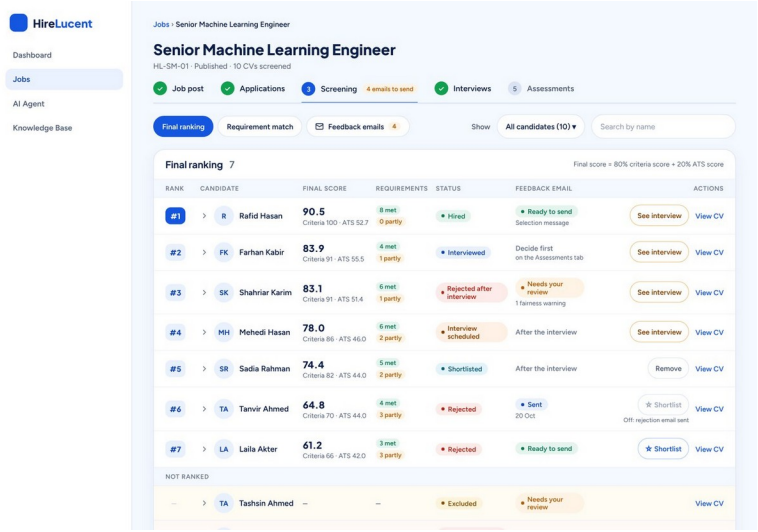


Figure 9: *Screening tab: final ranking, F1 and F2 (HL_UI_009)*

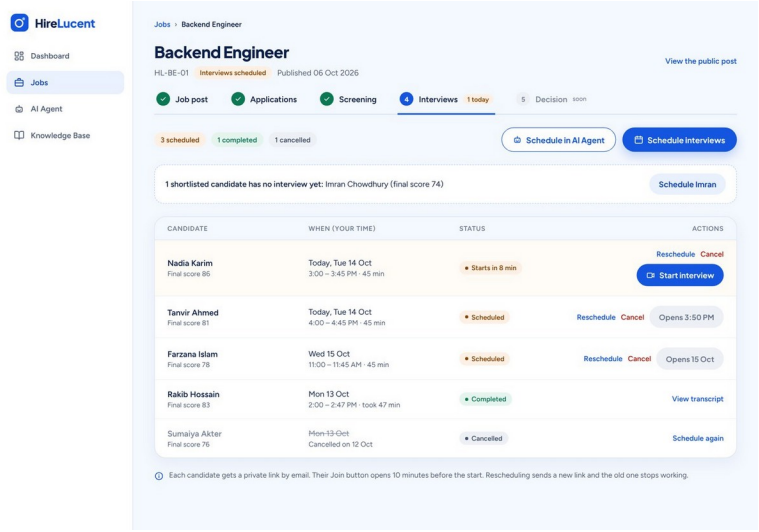


Figure 10: *Interviews tab: scheduled and completed interviews (HL_UI_010)*

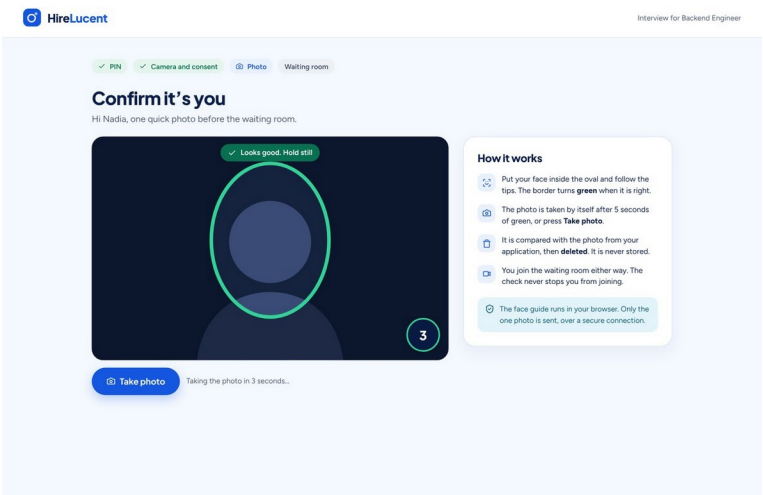


Figure 11: *Candidate identity photo, F7 (HL_UI_025)*

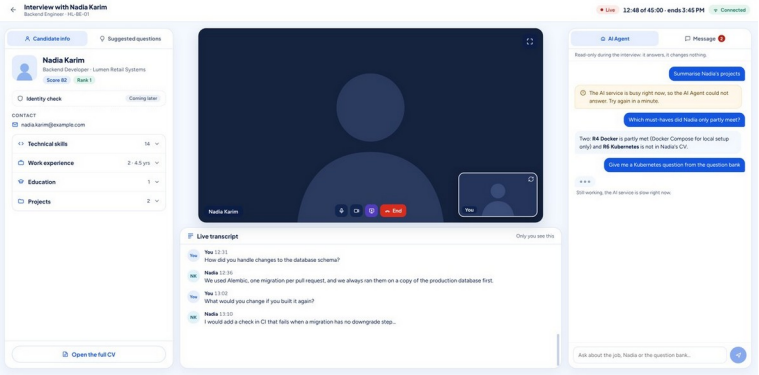


Figure 12: *Interviewer room with live transcript, F3 and F4 (HL_UI_012)*

Transcript, analysis and the assessment report

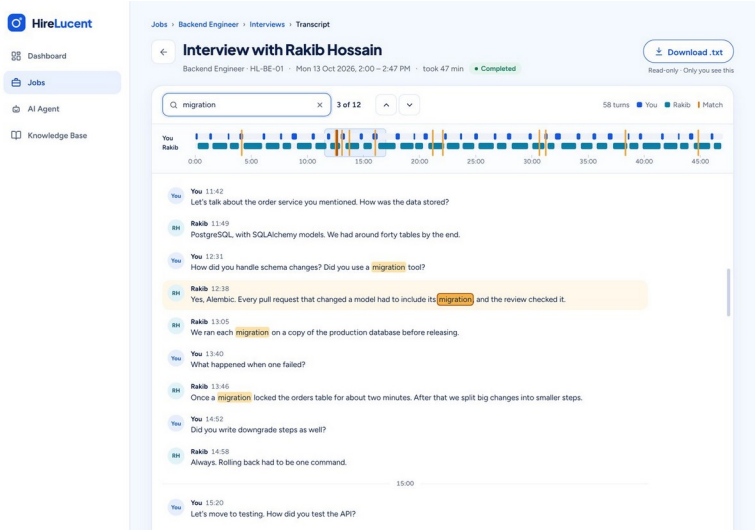


Figure 13: *Transcript viewer, F4 (HL_UI_013)*

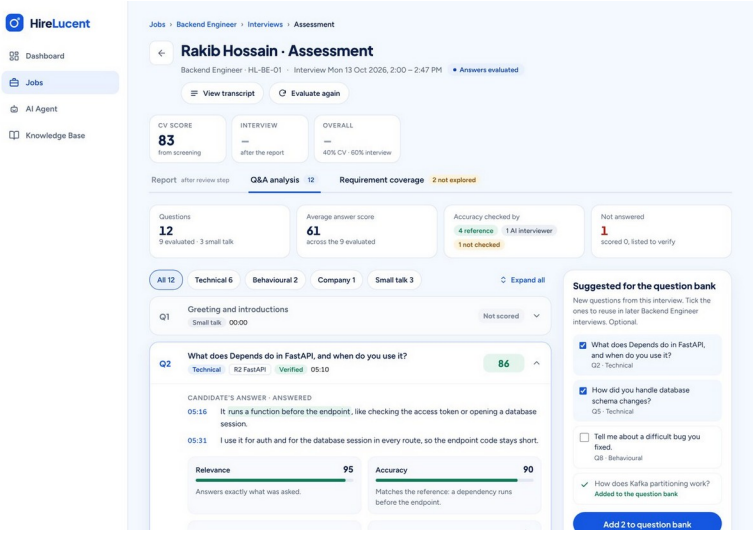


Figure 14: *Q&A analysis tab, F5 (HL_UI_016)*

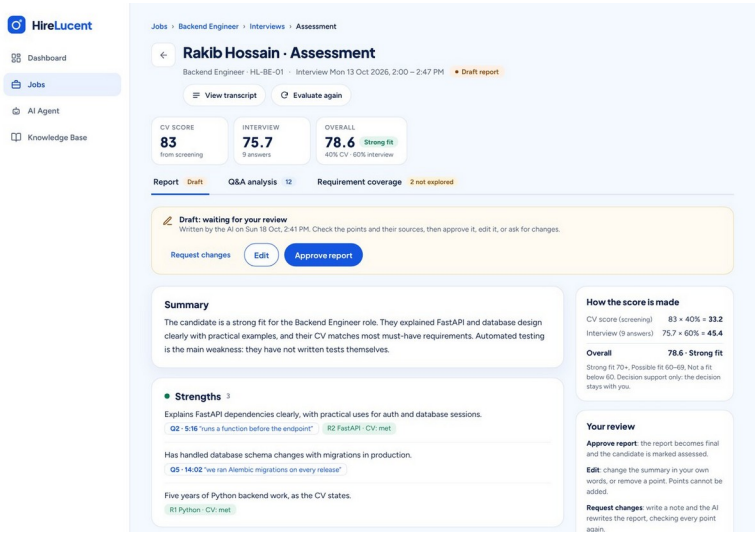


Figure 15: *Assessment report with review, F6 (HL_UI_014)*

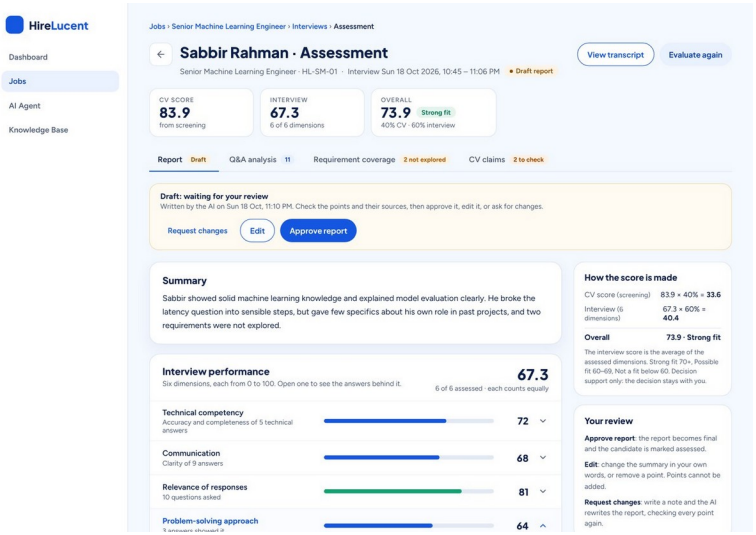


Figure 16: *Interview performance scoring, F9 (HL_UI_015)*

CV claims, interview quality, comparison and feedback

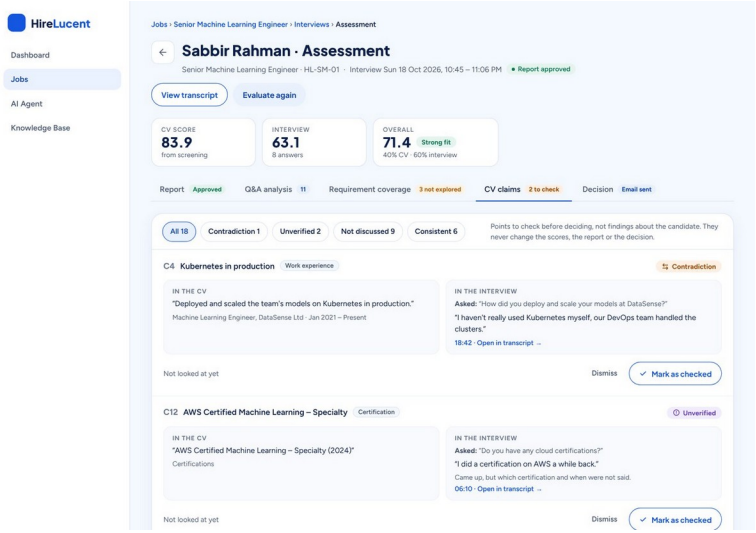


Figure 17: *CV claims tab, F8 (HL_UI_018)*

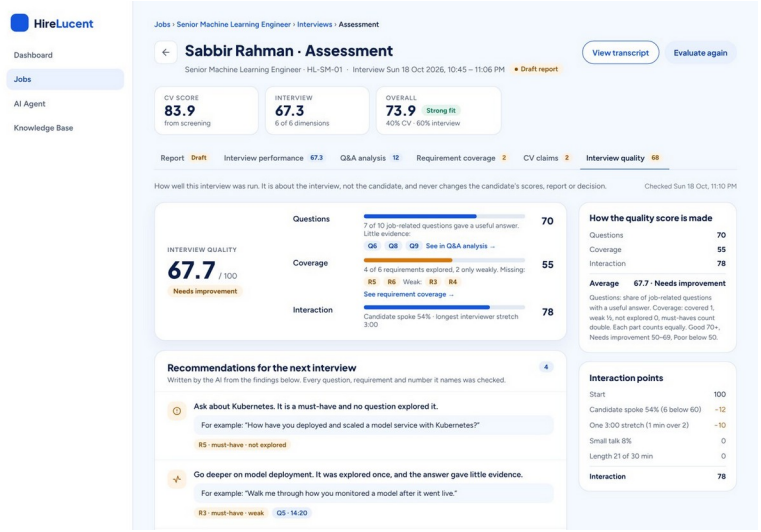


Figure 18: *Interview quality tab, F10 (HL_UI_019)*

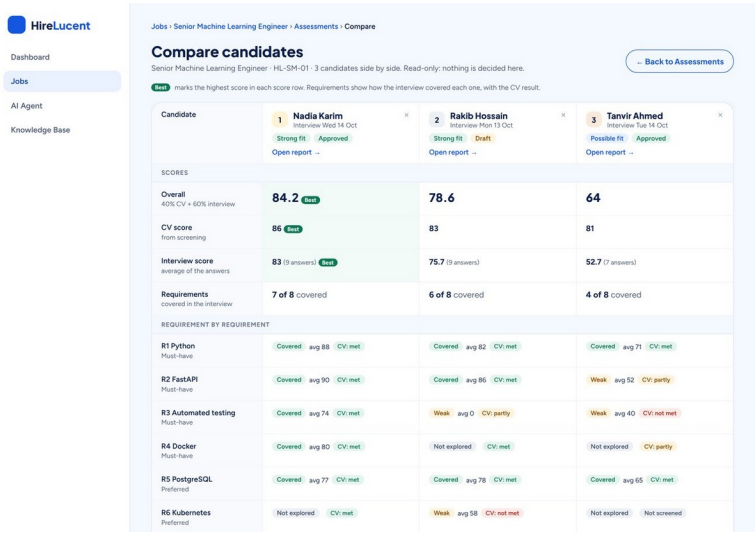


Figure 19: *Compare candidates, F6 (HL_UI_022)*

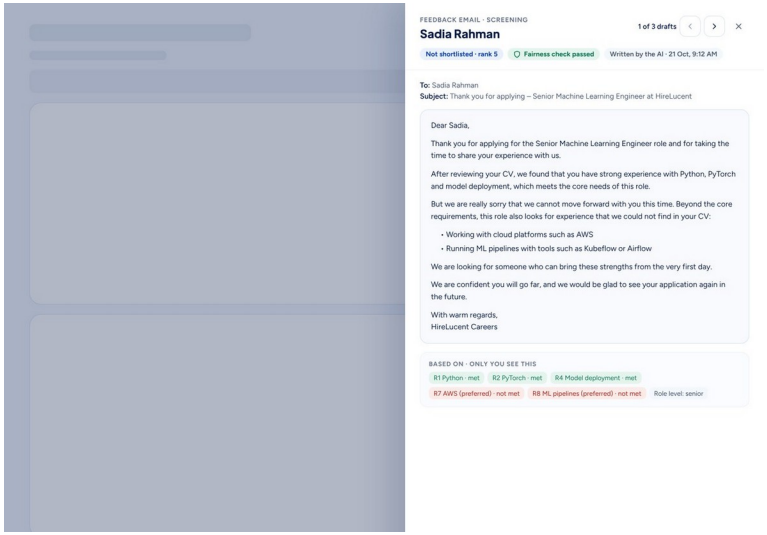


Figure 20: *Feedback email review panel, F11 (HL_UI_021)*

5. Non-Functional Requirements

This section states the quality levels HireLucent must meet. General requirements are listed first, followed by performance, safety and security requirements. They are derived from BID Section 7 (N1–N26) and the agreed clarifications.

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_NFR_001	The application shall start on a clean machine with one documented command (docker compose up -d --build).	Startup succeeds with no undocumented manual steps.	Docker Desktop required; Google sign-in prepared as documented.	TC_NFR_001
HL_NFR_002	Candidate records, scores, transcripts, assessments and AI workflow state shall survive a restart.	Data is present after restarting all containers; interrupted tasks are marked failed and can be run again.	Named Docker volumes.	TC_NFR_002
HL_NFR_003	Long-running AI operations shall run in the background with progress or status feedback.	No request blocks the interface; progress cards are shown.	—	TC_NFR_003
HL_NFR_004	Failure of an external AI service shall produce a clear error state, never a crash or silent data loss.	A readable message is shown and the operation can be repeated.	Up to three attempts per AI call.	TC_NFR_004
HL_NFR_005	Key operations and failures shall be logged as structured JSON with identifiers and counts only.	Logs contain no names, emails, CV text, transcripts, prompts, tokens or keys.	—	TC_NFR_005
HL_NFR_006	The system shall operate in English.	All screens, emails and analysis are in English.	Clarifications #29 and #44.	TC_NFR_006

5.1 Performance Requirements

The performance characteristics below are the targets agreed for HireLucent on the reference demonstration PC (8 GB RAM, Docker Desktop). AI-dependent timings are bounded by the free-tier limits of the language model service.

- **Response time:** non-AI API requests respond within one second under normal load; AI operations are asynchronous and report progress.
- **CV screening throughput:** a batch of 20 CVs is parsed, screened, scored and ranked within about 5 minutes (clarification #23); the measured figure is recorded in the README after the end-to-end test.
- **Transcription latency:** live words appear while being spoken and saved lines arrive within 2 seconds of the speaker stopping. Measured on 17 October 2026 with AssemblyAI: 0.08–0.09 seconds on average and under 0.9 seconds for 95% of lines.
- **Capacity and concurrency:** one live interview at the same time as background CV screening or evaluation; up to 4 CVs screened in parallel.

- **AI request pacing:** at most 14 language model requests per minute overall, of which background work may use 10, so the interviewer's chat stays responsive.
- **Degradation modes:** if the AI service is unavailable, screening and evaluation stop with a clear message and can be repeated; if speech-to-text drops, audio is queued for 30 seconds and the stream reconnects; if the network drops, the video reconnects automatically while the transcript saved so far is kept.
- **Resource utilisation:** the full stack runs within the memory available to Docker on an 8 GB PC; raw audio and video are never written to disk.

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_PERF_001	A batch of 20 CVs shall be screened, scored and ranked within about 5 minutes.	Measured time from Screen to ranking recorded in the README.	Depends on AI service limits.	TC_PERF_001
HL_PERF_002	Saved transcript lines shall appear within 2 seconds of the speaker stopping.	Average and 95th percentile delay logged per stream.	Depends on the speech-to-text provider.	TC_PERF_002
HL_PERF_003	One live interview shall run stably alongside background screening or evaluation.	No dropped session or failed task during a combined run.	Single concurrent interview (BID A9).	TC_PERF_003

5.2 Safety Requirements

These requirements protect candidates and the hiring organisation from harm that could result from automated decisions, unfair treatment or unintended external actions.

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_SAF_001	Every external or consequential action (saving or publishing a job, scheduling, rescheduling or cancelling an interview, sending a feedback email) shall wait for the interviewer's explicit approval.	No external effect occurs without an approval record.	External tools are never offered to the language model directly.	TC_SAF_001
HL_SAF_002	AI results shall be decision support only; the hiring decision shall always be made by the interviewer.	No candidate is hired, rejected after interview or emailed automatically.	Must-have exclusion at screening is a defined rule (clarification #9).	TC_SAF_002
HL_SAF_003	Every score, flag and report statement shall be traceable to CV content or a transcript segment.	Each point shows its evidence or is removed.	—	TC_SAF_003

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_SAF_004	Scoring shall use job-related criteria only and never protected characteristics such as age, gender, nationality, religion or accent.	Fairness instructions and inputs verified by the automated fairness test.	Text may still mention personal details; documented in the README.	TC_SAF_004
HL_SAF_005	CV claim flags and identity results shall be advisory and never accusatory or blocking.	Neutral wording is enforced; joining is never blocked by the face check.	—	TC_SAF_005
HL_SAF_006	A candidate with an upcoming or running interview shall not be deletable until the interview is cancelled.	Deletion is refused with a clear message.	Prevents a live invitation without a candidate record.	TC_SAF_006

5.3 Security Requirements

The following factors protect HireLucent and its data from accidental or malicious access, use, modification, destruction or disclosure:

- Secrets supplied only through environment variables; a .env.example with placeholders is shipped and .env and secrets/ are never committed.
- Passwords, interview PINs and join link tokens stored only as hashes (Argon2 and SHA-256).
- Signed, short-lived tokens for each role and each interview room.
- Interviewer-only data separated from candidate capabilities and verified for every route.
- Inputs, file types and sizes validated before processing; error responses never reveal internals.
- Structured logs without personal data; dependency versions pinned.

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_SEC_001	All API keys, secrets and connection strings shall come from environment variables; no secret shall be committed.	.env.example contains placeholders only; .env and secrets/ are git-ignored.	—	TC_SEC_001
HL_SEC_002	Interviewer passwords shall be hashed with Argon2; interview PINs with Argon2; join link tokens with SHA-256.	Only hashes are present in the database.	—	TC_SEC_002
HL_SEC_003	Five wrong PIN attempts shall lock the interview link for 15 minutes.	The sixth attempt within the window is refused.	—	TC_SEC_003
HL_SEC_004	All API inputs shall be validated by Pydantic models with size limits; uploads shall be checked for type and size.	Invalid input returns 422 or 413 with a readable message.	CV and document uploads up to 10 MB.	TC_SEC_004

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_SEC_005	Candidates shall never access scores, rankings, reports or transcripts.	Access-separation test passes for every route.	—	TC_SEC_005
HL_SEC_006	Error responses shall not include stack traces, file paths, configuration or prompt content.	Clients receive a generic message; details stay in the logs as error types and code locations.	—	TC_SEC_006
HL_SEC_007	Calls to external services shall use HTTPS or secure WebSockets.	Gemini, Google, Deepgram and AssemblyAI use secure transport.	The local app runs over HTTP and LiveKit over ws:// on localhost (documented exception).	TC_SEC_007
HL_SEC_008	Candidate contact details shall be masked in every text sent to the language model.	Prompts contain no email addresses, phone numbers or links.	—	TC_SEC_008
HL_SEC_009	Biometric data shall be limited to the re-saved application photo and its face code, used only for the identity check and deleted with the candidate or job.	No face image leaves the PC; the joining photo and camera frames are never stored.	Consent required at application and at joining.	TC_SEC_009
HL_SEC_010	Backend dependency versions shall be pinned; frontend versions locked with package-lock.json.	requirements.txt uses exact versions; npm ci reproduces the frontend.	—	TC_SEC_010

6. Design Constraints

The following design decisions are mandated by the BID, the approved clarifications or the client change requests, and must be adhered to.

- **Mandated stack:** all backend and AI logic in Python with FastAPI and Pydantic; agent workflows modelled as LangGraph graphs with explicit state, nodes, edges, retries and checkpoints; the complete application containerised with Docker Compose (BID 8.1).
- **Meaningful LangGraph use:** six multi-node workflows (Recruiter Agent, CV Screening, Q&A Analysis, Final Assessment, Candidate Feedback, Interview Quality); no single-call graphs (BID 8.2).
- **Architecture:** service-oriented backend (routes, services, graphs, schemas, models) with tool access through MCP servers; external tools are called by the backend only after approval and are never listed to the language model.
- **Data stores:** PostgreSQL for records and LangGraph checkpoints, ChromaDB for vectors; database changes only through Alembic migrations.
- **Deployment:** local execution with named volumes and a single startup command; no cloud services for hosting.
- **Third-party components:** LiveKit (video), Deepgram or AssemblyAI (speech-to-text), Gemini (language model), Google Workspace and Blogger APIs; open-source face models YuNet (MIT), SFace (Apache 2.0) and MediaPipe (Apache 2.0); community Blogger MCP server (MIT).
- **Roles and access:** two roles only, with token-based access; no enterprise identity integration (BID 4.2).
- **Data policy:** sample or synthetic data only; raw audio and video never stored; English only.
- **Source control:** GitHub with main, dev and one branch per module, merged by pull request; meaningful, incremental commits.

7. Software Quality Attributes

The quality attributes below describe how well HireLucent must perform its functions over time.

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_QA_001	Reliability: AI workflows shall retry failed or invalid AI output and keep only results that pass their checks.	Up to three attempts; untrusted output is dropped, never kept without evidence.	—	TC_QA_001
HL_QA_002	Resilience: an interrupted task shall be recoverable without corrupting saved results.	Tasks interrupted by a restart are marked failed and can be started again.	—	TC_QA_002
HL_QA_003	Testability: core business logic shall be covered by automated tests.	More than 1,000 backend tests pass, including access-separation, fairness, deletion and input-limit tests.	Frontend checked by lint and build.	TC_QA_003
HL_QA_004	Maintainability: code shall follow a consistent structure, clear naming and separation of concerns with no dead code.	Lint and review pass; modules follow routes / services / graphs / schemas.	—	TC_QA_004
HL_QA_005	Usability: the interface shall be usable without verbal explanation; long operations show progress and errors are understandable.	A first-time interviewer completes the workflow using on-screen guidance.	—	TC_QA_005
HL_QA_006	Explainability: every AI result shall show the evidence and rules behind it.	Scores, flags and recommendations link to their sources.	—	TC_QA_006
HL_QA_007	Portability: the system shall run on any machine with Docker Desktop.	Starts from a fresh clone on a Windows PC with Docker Desktop using the README commands.	Windows with WSL 2 is the reference platform.	TC_QA_007

8. User Interface (General Guidelines)

Section 4 lists the individual screens. This section states the user interface rules that apply across all of them.

UI No.	UI Name	Related Info No.	Description	Notes	Test case Identifier
HL_UIG_001	Visual style	HL_UI_001 – 028	Clean corporate design: white and pale-blue backgrounds, blue primary colour, navy headings, rounded cards.	One colour per hiring stage.	TC_UIG_001
HL_UIG_002	Progress feedback	HL_NFR_003	Every long action shows a progress card; every short action a busy button.	—	TC_UIG_002
HL_UIG_003	Error messages	HL_NFR_004, HL_SEC_006	Errors show a plain sentence that tells the user what to do next.	Fallback messages for large files, server and network problems.	TC_UIG_003
HL_UIG_004	Confirmation dialogs	HL_SAF_001	Consequential actions (approve, send, delete, reschedule, cancel) ask for confirmation.	Shared confirm dialog with busy state.	TC_UIG_004
HL_UIG_005	Time display	HL_SCH_002, HL_F4_003	Times are stored in UTC and shown in the browser’s local time.	Typed times read in the configured time zone.	TC_UIG_005
HL_UIG_006	Evidence links	HL_SAF_003	Quotes and findings link to the transcript turn or report section they come from.	—	TC_UIG_006
HL_UIG_007	Status indicators	HL_UI_007 – 011	Job tabs show a tick when their step is finished and badges for pending work.	—	TC_UIG_007
HL_UIG_008	Screen sizes and motion	—	Layouts work from laptop width (1280 px) upward; animations respect the reduced-motion setting.	Phone layout is not a target.	TC_UIG_008

9. Other Requirements

This section defines requirements not covered elsewhere: database, data retention, legal and privacy, documentation, budget and schedule.

Requirement ID	Requirement Description	Acceptability / Completion Criteria	Limitations / Constraints	Test Case Identifier
HL_OTH_001	Database: all persistent data shall be stored in PostgreSQL with schema changes through Alembic migrations applied at startup.	A fresh start applies every migration automatically.	—	TC_OTH_001
HL_OTH_002	Data retention: AI working state shall be kept no longer than needed; Q&A analysis state is deleted when each run ends.	No analysis checkpoint remains after a run.	—	TC_OTH_002
HL_OTH_003	Third-party disclosure: the README shall list every external service that receives candidate data and what it receives.	Gemini, Deepgram or AssemblyAI, Google Calendar and Gmail are listed.	AssemblyAI free plan cannot opt out of model training; test use only.	TC_OTH_003
HL_OTH_004	Legal and privacy: biometric processing shall require explicit consent and be documented with storage location and retention.	Consent is visible in the form and before joining.	—	TC_OTH_004
HL_OTH_005	Documentation: the README shall cover purpose, architecture, setup, configuration, usage, privacy, fairness and known limitations; the API shall be documented by OpenAPI.	README sections present; /docs lists every endpoint.	—	TC_OTH_005
HL_OTH_006	Budget: AI usage cost shall be measurable per CV, interview, job and hire.	The AI cost report produces these unit costs.	Free tier used in development; costs shown as production estimates.	TC_OTH_006
HL_OTH_007	Schedule: all documents and code shall be submitted by 27 October 2026 (clarification #13).	Final submission tagged in the repository.	Fixed date.	TC_OTH_007

9.1 Requirements Traceability

The matrix below traces each BID feature and non-functional requirement group to the requirements in this specification, so that every business need can be followed to its implementation and test.

BID item	SRS requirement identifiers
F1 – CV Screening & Ranking	HL_F1_001 – 007
F2 – AI-Based Candidate Scoring	HL_F2_001 – 006
F3 – AI-Powered Video Interview	HL_F3_001 – 007, HL_SCH_001 – 005
F4 – Speech Recognition & Transcription	HL_F4_001 – 007, HL_PERF_002
F5 – Interview Q&A Analysis	HL_F5_001 – 006, HL_KB_001 – 004
F6 – Consolidated Candidate Assessment	HL_F6_001 – 007
F7 – Candidate Identification (Stretch)	HL_F7_001 – 005, HL_SEC_009
F8 – CV Claim Verification (Stretch)	HL_F8_001 – 005
F9 – Interview Scoring (Stretch)	HL_F9_001 – 005
F10 – Interview Quality (Stretch)	HL_F10_001 – 005
F11 – Innovation: Feedback Composer	HL_F11_001 – 005
Client change: job post and Google Form	HL_JOB_001 – 007, HL_PUB_001 – 005
N1 – N6 Performance & responsiveness	HL_PERF_001 – 003, HL_NFR_003 – 004
N7 – N12 Security	HL_SEC_001 – 010, HL_AUTH_001 – 005
N13 – N20 Candidate data privacy	HL_SEC_008 – 009, HL_SAF_004, HL_OPS_001 – 002, HL_OTH_002 – 004
N21 – N26 Reliability, usability, persistence	HL_NFR_001 – 005, HL_QA_001 – 007

End of document.